

Project Proposal

1. Project Description

Our team proposes building a robotic system capable of playing a physical game of Monopoly against human opponents. To make the project possible within the given timeframe and to avoid the difficulties of robotic manipulation of thin paper (money and property cards), we propose a "mixed-reality" approach. The game will use a real, physical game board, characters, houses and dice. However, the "Bank" will be entirely digital, managed via a UI.

The robot will physically roll the dice, read the outcome via an overhead camera, calculate its move, and physically pick up and place its character on the correct space. By eliminating paper handling, we can focus on precise manipulation, computer vision and game logic.

2. Hardware Requirements

- **Manipulator:** UR5e
- **End-Effector:** a gripper or some sort of "hand"(for grasping characters, houses, and dice).
- **Vision:** Overhead Web Camera (mounted above the workspace).
- **Props:** Physical Monopoly board, dice, and customized/3D-printed characters and houses.
- **Interface:** A monitor or tablet to display the UI.
- **PC**

3. Software Architecture

The software will rely on the Simple Manipulator Control (SMC) library and be divided into four main pillars:

- **Control & Kinematics (SMC):** Inverse Kinematics to calculate precise drop-off locations. We will implement collision-free trajectory planning.
- **Computer Vision:** Using some CV library to process the overhead camera feed, detect dice values, and track characters.

- **Game Engine & AI:** A Python or MATLAB-based state machine that maintains the rules and decision-making for the robot's financial strategy.
- **UI & System Integration:** A Digital Bank interface to connect vision, game engine, and control.

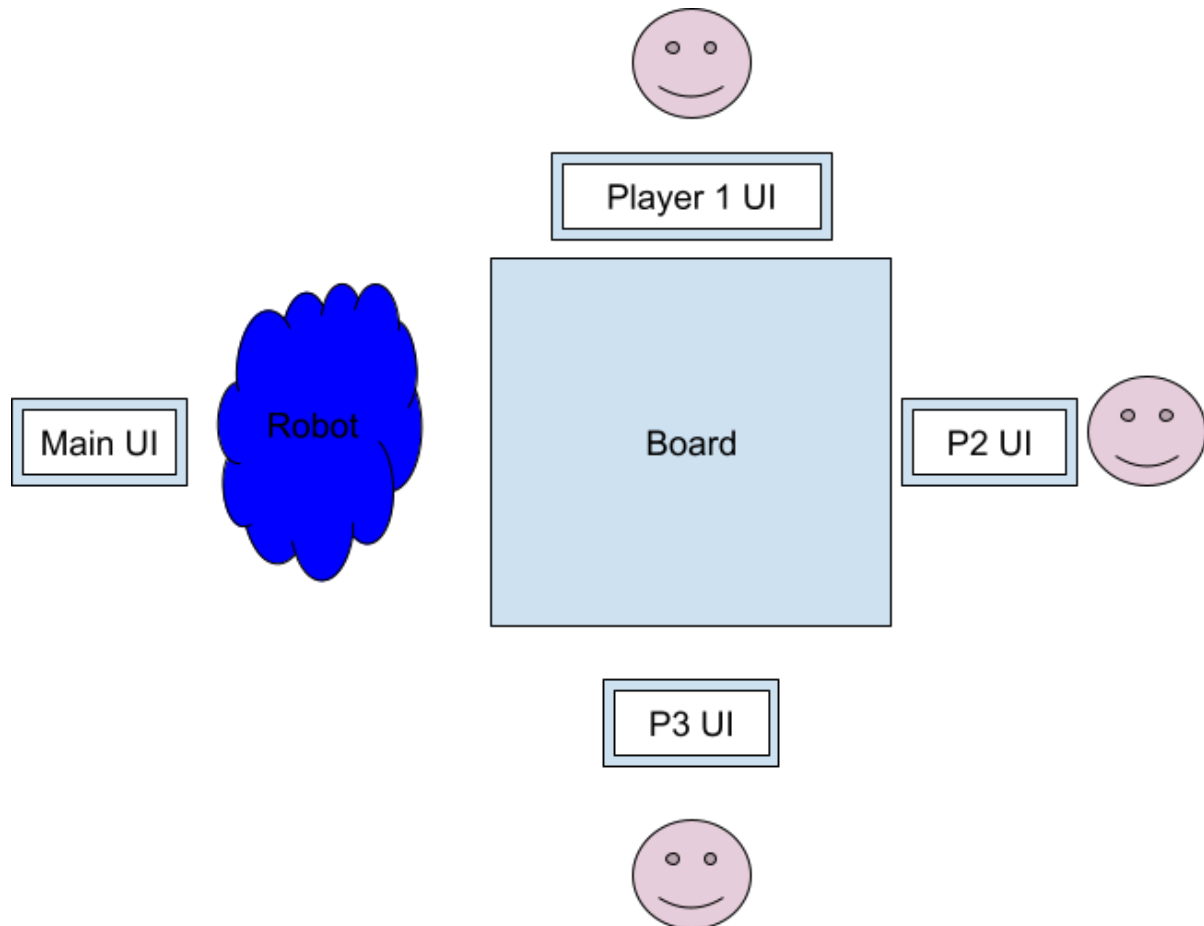
4. Task Distribution

- **Vision Sub-team (2 members):** Camera calibration, dice recognition, and token tracking.
- **Control Sub-team (2 members):** Integration with the SMC library, IK solvers, trajectory generation, and gripper actuation.
- **AI & Logic Sub-team (2 members):** Programming the Monopoly game engine, managing digital states, and developing strategy.
- **Integration & UI Lead (1 member):** Developing the Digital Bank GUI and ensuring seamless communication across modules.

5. Fallback Plan

- **Tier 1 Simplification:** Remove computer vision. Humans input dice rolls into the UI.
- **Tier 2 Simplification:** The robot acts solely as the "Dice Roller" and "Banker". Humans move all physical pieces.

We imagine this would be the look of the project:



We could also have just one main UI where every player must select actions at every turn and the robot is able to show its current state.

Some slight problems to consider:

- Having a polyglot system with Matlab and Python adds latency to the game and complexity to the main process. But using the matlab library gives us a fully implemented game engine which would take us a lot of time to build.
- Having a main process that runs the matlab game engine, the python robot commands and monitoring, CV + players UI is not lightweight, so we would need a good PC in order to prevent the process to crash